

A unified kinetic flow rule for body-centered-cubic metals: derivation from the mean dislocation velocity to complete constitutive equations for W, Zr and Eurofer-97

Matthew Maron, Giacomo Po, Yang Li

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1 Starting point: the mean velocity of a dislocation segment

Plastic shear is carried by mobile dislocations of density ρ_m moving with a mean velocity $\bar{v}$ (Orowan [1]):

$$\dot{\gamma} = \rho_m b \bar{v}. \quad (1.1)$$

The whole model is contained in how $\bar{v}$ is written. A segment does not move at constant speed. Over one characteristic glide distance λ it spends a time t_r *travelling* through the lattice and a time t_w *waiting* at obstacles that it cannot pass immediately. The mean velocity is distance over total time,

$$\bar{v} = \frac{\lambda}{t_r + t_w} \quad (1.2)$$

and therefore

$$\frac{1}{\dot{\gamma}} = \frac{t_r + t_w}{\rho_m b \lambda}. \quad (1.3)$$

1.1 Travel time

Between obstacles the segment glides with the lattice velocity $v_g(\tau^*)$, which depends on the effective stress τ^* acting on the lattice-friction process. The travel time over λ is $t_r = \lambda/v_g$, so the travel term in Eq. (1.3) is

$$\frac{t_r}{\rho_m b \lambda} = \frac{1}{\rho_m b v_g(\tau^*)} \equiv \frac{1}{\dot{\gamma}_g}. \quad (1.4)$$

$\dot{\gamma}_g$ is the rate the material would have if travel were the only process.

1.2 Waiting time

Let obstacle populations j have areal densities λ_j^{-2} on the slip plane. A line that sweeps a strip of unit length meets n_j obstacles of type j , with

$$n_j = \lambda_j^{-2} \lambda, \quad \lambda^{-2} = \sum_j \lambda_j^{-2}, \quad (1.5)$$

where λ is the combined spacing. If an obstacle of type j is passed at a rate ν_j (s^{-1}), the mean time spent at it is $1/\nu_j$. The waiting time per unit glide distance is $\sum_j n_j/\nu_j$, so

$$\frac{t_w}{\rho_m b \lambda} = \frac{1}{\rho_m b} \sum_j \frac{n_j}{\nu_j} \equiv \frac{1}{\dot{\gamma}_w}. \quad (1.6)$$

1.3 Several ways past one obstacle

An obstacle may be passed in more than one way: thermally activated cutting or detachment, climb, cross slip, and so on. These routes are *alternatives*; whichever succeeds first releases the segment. The release rate is the sum of the route rates,

$$\nu_j = \sum_k \nu_{jk}(\tau^*). \quad (1.7)$$

1.4 The general flow rule

Combining the pieces gives

$$\boxed{\frac{1}{\dot{\gamma}} = \frac{1}{\rho_m b v_g(\tau^*)} + \frac{1}{\rho_m b} \sum_j \frac{n_j}{\sum_k \nu_{jk}(\tau^*)}} \quad (1.8)$$

Three composition rules are contained in this expression.

S (series). Processes a segment goes through one after another add their *times*: $\dot{\gamma}^{-1} = \sum_i \dot{\gamma}_i^{-1}$. The slowest step controls, but every step contributes.

P (parallel). Alternative routes past the same obstacle add their *rates*, Eq. (1.7). The fastest route controls.

L (linear superposition). Resistances that act on the line *at the same time* add their *stresses*. This is the case for long-range internal stresses (the Taylor forest stress, boundary stresses) and for sparse strong obstacles that pin a line which is simultaneously held back by lattice friction [2, 3]:

$$\tau = \tau_a + \tau^* + \sum_g \tau_g. \quad (1.9)$$

1.5 Which obstacles enter where

The choice between rules S and L follows from the obstacle strength and spacing.

- *Forest junctions* in BCC metals are strong and long range. They are treated as an athermal back stress τ_a (rule L), not as waiting sites. Their rate dependence enters through the forest *density*, which evolves with rate and temperature (Sec. 4).
- *Carbides and carbonitrides* ($M_{23}C_6$, MX) in Eurofer-97 are sparse and much stronger than the lattice friction. The line bows between them while the lattice friction acts along its whole length, so their stress is superposed (rule L). Each such group, however, has its own *time* balance. A segment pinned at particles of spacing λ_p is released at a rate $\nu_p = \sum_k \nu_{pk}(\tau_p)$ (rule P), and between releases it advances by λ_p . Applying Eq. (1.8) to this group alone gives

$$\dot{\gamma} = \frac{\rho_m b}{n_p / \nu_p} = \rho_m b \lambda_p \nu_p(\tau_p), \quad n_p = \lambda_p^{-1}, \quad (1.10)$$

which fixes the particle stress τ_p for the imposed rate.

- *Weak, dense, thermally penetrable obstacles* (e.g. small clusters or irradiation loops) would enter the waiting sum of Eq. (1.8) (rule S). None of the three materials treated here needs such a population, so for them the waiting sum over intragranular obstacles is zero and Eq. (1.8) reduces to

$$\dot{\gamma} = \rho_m b v_g(\tau^*). \quad (1.11)$$

Diffusional (Nabarro–Herring and Coble) flow transports matter rather than dislocations. It is an independent carrier, so its rate adds to the dislocation rate (Sec. 6).

2 Travel: kink-pair glide

Screw dislocations in BCC metals, and on prismatic planes in Zr, move by the thermally activated nucleation of kink pairs followed by lateral kink migration [4, 5]. We use the mobility law of Po *et al.* [6]:

$$v_g(\tau^*, T) = \frac{\tau^* b}{B_d} \exp \left\{ -\frac{\Delta H_0}{2k_B T} \max \left[\left(1 - \left(\frac{\tau^*}{\tau_P} \right)^p \right)^q - \frac{T}{T_0}, 0 \right] \right\} \quad (2.1)$$

The terms are as follows.

- $\tau^* b/B_d$ is the viscous drag velocity, with drag coefficient B_d .
- $\Delta H_0[1 - (\tau^*/\tau_P)^p]^q$ is the kink-pair enthalpy in the Kocks–Argon–Ashby form [2]; it vanishes at the Peierls stress τ_P .
- The factor $1/2$ applies to long screw segments, where the rate is set by the square root of the nucleation rate times the kink migration rate.
- $-T/T_0$ lowers the barrier linearly with temperature and acts as an entropic correction. The $\max[\cdot, 0]$ ensures that above the temperature where the barrier vanishes, glide becomes drag controlled.

For prismatic slip in Zr the drag coefficient switches smoothly from the kink value B_k to the free-flight value B_0 as the barrier $\Delta g = \max[\dots]$ disappears:

$$B_d = B_k(1 - s) + B_0 s, \quad s = \left[1 + \exp \left(-\frac{0.05 - \Delta g}{0.05} \right) \right]^{-1}. \quad (2.2)$$

The **intrinsic mobility sensitivity**, used below, is

$$A \equiv \frac{\partial \ln v_g}{\partial \tau^*} = \frac{1}{\tau^*} + \frac{\Delta H_0 p q}{2k_B T \tau_P} x^{p-1} (1 - x^p)^{q-1}, \quad x = \frac{\tau^*}{\tau_P}, \quad (2.3)$$

where the second term applies while the barrier is non-zero.

3 Waiting: routes past sparse strong obstacles

3.1 Thermally activated detachment or cutting

$$\nu_{th} = \nu_0 \exp \left\{ -\frac{F}{k_B T} \left[1 - \left(\frac{\tau_p}{\hat{\tau}} \right)^p \right]^q \right\}, \quad (3.1)$$

where F is the activation free energy at zero stress and $\hat{\tau}$ is the athermal (Orowan) strength [2, 7]. At $\tau_p \geq \hat{\tau}$ the barrier vanishes and $\nu_{th} = \nu_0$. Its sensitivity is

$$A_{th} = \frac{\partial \ln \nu_{th}}{\partial \tau_p} = \frac{F p q}{k_B T \hat{\tau}} y^{p-1} (1 - y^p)^{q-1}, \quad y = \frac{\tau_p}{\hat{\tau}}. \quad (3.2)$$

3.2 Glide-assisted local climb

A segment held at a particle can escape by climbing over it. The climb velocity of an edge segment under a climb stress τ_p follows from the vacancy flux to a cylindrical core of radius r_c from a distance R [4]:

$$v_c = \frac{2\pi D_c}{b \ln(R/r_c)} \left[\exp\left(\frac{\tau_p \Omega}{k_B T}\right) - 1 \right]. \quad (3.3)$$

The segment must climb a height h_{eff} . In local climb the line is pressed against the particle, and the part that still has to climb shrinks as the applied stress approaches the particle strength [8, 9]. We write

$$h_{\text{eff}} = h \max\left(1 - \frac{\tau_p}{\hat{\tau}}, 10^{-3}\right), \quad (3.4)$$

so that

$$\nu_{\text{cl}} = \frac{v_c}{h_{\text{eff}}} = \frac{2\pi D_c}{b h \ln(R/r_c)} \frac{e^{\tau_p \Omega / k_B T} - 1}{1 - \tau_p / \hat{\tau}}, \quad (3.5)$$

with sensitivity

$$A_{\text{cl}} = \frac{\partial \ln \nu_{\text{cl}}}{\partial \tau_p} = \frac{\Omega}{k_B T} \frac{e^u}{e^u - 1} + \frac{1}{\hat{\tau} - \tau_p}, \quad u = \frac{\tau_p \Omega}{k_B T}. \quad (3.6)$$

3.3 Combined release rate and the particle stress

By rule P, $\nu_p = \nu_{\text{th}} + \nu_{\text{cl}}$, and Eq. (1.10) gives the implicit equation for τ_p :

$$\boxed{\dot{\gamma} = \rho_m b \lambda_p [\nu_{\text{th}}(\tau_p) + \nu_{\text{cl}}(\tau_p)]} \quad (3.7)$$

Both rates increase with τ_p , so the solution is unique. The sensitivity of the group is the rate-weighted mean

$$A_p = \frac{\nu_{\text{th}}}{\nu_p} A_{\text{th}} + \frac{\nu_{\text{cl}}}{\nu_p} A_{\text{cl}}, \quad \left. \frac{d\tau_p}{d \ln \dot{\gamma}} \right|_{\rho_m} = \frac{1}{A_p}. \quad (3.8)$$

Threshold creep. When climb controls and $u \ll 1$, $\nu_{\text{cl}} \propto \tau_p / (1 - \tau_p / \hat{\tau})$, and the local stress exponent of the particle process is

$$n_p \equiv \frac{\partial \ln \nu_{\text{cl}}}{\partial \ln \tau_p} = \tau_p A_{\text{cl}} \simeq \frac{1}{1 - \tau_p / \hat{\tau}}. \quad (3.9)$$

This exponent diverges as $\tau_p \rightarrow \hat{\tau}$, which is the origin of the large and temperature-dependent Norton exponents of particle-strengthened steels. As $D_c(T)$ increases, a given rate is reached at a smaller $\tau_p / \hat{\tau}$, and n_p falls.

4 State: dislocation density and athermal stress

4.1 Taylor stress

The long-range athermal stress is [10, 11]

$$\boxed{\tau_a = \alpha \mu(T) b s_c (\sqrt{\rho_f} + \Lambda_b)} \quad (4.1)$$

where $\rho_f = \beta \rho$ is the forest part of the total density ρ , Λ_b is the inverse spacing of boundaries (laths, blocks, grains, particles), and s_c is the solute strengthening factor of Sec. 5 ($s_c = 1$ without aging). The mobile density is $\rho_m = f_m \rho_f$.

4.2 Evolution of the dislocation density

Three processes change ρ during deformation.

Storage. A mobile dislocation that glides a distance dx is stored after a mean free path L , so $d\rho = d\gamma/(bL)$. With $1/L \propto \sqrt{\rho}$ (forest) plus $1/L \propto 1/d_i$ (boundaries) this gives $k_1(\sqrt{\rho} + \Lambda_b)$ [12, 13, 14, 15].

Dynamic recovery. Moving dislocations annihilate stored ones they pass within a capture distance, mainly by cross slip. The number annihilated per unit strain is proportional to ρ , giving $-k_2\rho$ with $k_2 = k_{20} \exp(q_2/k_B T)$ [12, 13]. This process acts per unit *strain*.

Climb-controlled (static) recovery. Dipoles and network links annihilate by climb whether or not the crystal deforms. The classical rate per unit *time* is second order in density [16, 17, 18, 19]:

$$\left. \frac{d\rho}{dt} \right|_{\text{climb}} = -K D(T) \rho^2, \quad D = D_0 e^{-Q_D/k_B T}. \quad (4.2)$$

Dividing by $\dot{\gamma} = d\gamma/dt$ converts it to a per-strain rate.

Together,

$$\boxed{\frac{d\rho}{d\gamma} = k_1 (\sqrt{\rho} + \Lambda_b) - k_2 \rho - \frac{K D(T)}{\dot{\gamma}} \rho^2} \quad (4.3)$$

or, equivalently, in time form,

$$\frac{d\rho}{dt} = \dot{\gamma} [k_1 (\sqrt{\rho} + \Lambda_b) - k_2 \rho] - K D(T) \rho^2. \quad (4.4)$$

The same three-term structure is used by Sandström and co-workers for stress-strain curves and creep of copper [17, 20].

4.3 Steady state

The steady-state density ($d\rho/d\gamma = 0$) is the positive root of

$$\boxed{f(x) \equiv k_1(x + \Lambda_b) - k_2 x^2 - c_3 x^4 = 0, \quad x = \sqrt{\rho}, \quad c_3 = \frac{K D(T)}{\dot{\gamma}}} \quad (4.5)$$

Since $f(0) = k_1 \Lambda_b \geq 0$, $f''(x) = -2k_2 - 12c_3 x^2 < 0$ and $f \rightarrow -\infty$, there is exactly one positive root. It has two useful limits.

Low temperature ($c_3 \rightarrow 0$, $\Lambda_b = 0$): $x = k_1/k_2$, so

$$\tau_a = \alpha \mu b \sqrt{\beta} \frac{k_1}{k_2(T)}, \quad (4.6)$$

which is independent of rate.

Recovery controlled ($c_3 x^4 \gg k_2 x^2$, $x \gg \Lambda_b$): $k_1 x \simeq c_3 x^4$, so

$$\sqrt{\rho} = \left(\frac{k_1 \dot{\gamma}}{K D} \right)^{1/3} \implies \tau_a = \alpha \mu b \sqrt{\beta} \left(\frac{k_1 \dot{\gamma}}{K D} \right)^{1/3} \implies \dot{\gamma} = \frac{K D}{k_1} \left(\frac{\tau_a}{\alpha \mu b \sqrt{\beta}} \right)^3. \quad (4.7)$$

This is the natural creep law [21, 22]: stress exponent $n = 3$ and activation energy Q_D .

5 State: solute aging

5.1 Waiting time

A mobile segment is arrested at forest junctions of spacing $\lambda_f = \rho_f^{-1/2}$. Its mean velocity is $\bar{v} = \dot{\gamma}/(\rho_m b)$, so the time it waits between jumps is

$$t_a = \frac{\lambda_f}{\bar{v}} = \frac{\rho_m b \lambda_f}{\dot{\gamma}} \quad (5.1)$$

(cf. the waiting time $\Omega/\dot{\epsilon}$ of McCormick [23] and Kubin–Estrin [24]). It is fixed by the imposed rate and the state and does not depend on the stress.

5.2 Atmosphere formation

During t_a , solute species i diffuse to the arrested segment. Cottrell–Bilby kinetics give growth as $t^{2/3}$ [25], and saturation follows Louat [26]:

$$\boxed{\frac{C}{C_0} = 1 + \sum_i C_i [1 - e^{-x_i}], \quad x_i = \left(t_a \nu_i e^{-Q_i/k_B T}\right)^{\alpha_i}} \quad (5.2)$$

Here C_i is the saturation enrichment, ν_i an attempt frequency, Q_i the effective migration energy, and α_i the kinetic exponent (2/3 for Cottrell–Bilby, smaller when saturation is gradual).

5.3 Strengthening

Atmospheres increase the junction strength. We use [27]

$$s_c = \sqrt{C/C_0}, \quad (5.3)$$

which multiplies the Taylor stress in Eq. (4.1).

5.4 Rate dependence

Because $t_a \propto \dot{\gamma}^{-1}$ at fixed state,

$$\frac{\partial \ln(C/C_0)}{\partial \ln \dot{\gamma}} = - \frac{\sum_i C_i \alpha_i x_i e^{-x_i}}{C/C_0} < 0. \quad (5.4)$$

Faster straining leaves less time to age, so aging makes the resistance *decrease* with rate. The effect is largest when $x_i \approx 1$, i.e. when $t_a \approx e^{Q_i/k_B T}/\nu_i$; this condition places the DSA window in the $(T, \dot{\gamma})$ plane.

6 Diffusional flow

Nabarro–Herring and Coble creep [28, 29, 30] in the form of Frost and Ashby [31]:

$$\boxed{\dot{\gamma}_{\text{diff}} = 42 \frac{\Omega \tau}{k_B T d^2} \left(D_v + \frac{\pi \delta}{d} D_b \right)} \quad (6.1)$$

Here d is the grain size, $\delta = 2b$ the boundary width, and D_v and D_b the lattice and boundary diffusivities. The stress exponent is $n = 1$. The total rate is

$$\dot{\gamma}_{\text{tot}} = \dot{\gamma}_{\text{disl}}(\tau) + \dot{\gamma}_{\text{diff}}(\tau). \quad (6.2)$$

7 The assembled flow rule

Unified flow rule (all ingredients)

For an imposed shear rate $\dot{\gamma}$ and temperature T :

state:	$x = \sqrt{\rho} : k_1(x + \Lambda_b) - k_2x^2 - \frac{KD}{\dot{\gamma}}x^4 = 0, \quad \rho_f = \beta\rho, \quad \rho_m = f_m\rho_f,$
	$\frac{C}{C_0} = 1 + \sum_i C_i \left[1 - e^{-(t_a \nu_i e^{-Q_i/k_B T})^{\alpha_i}} \right], \quad t_a = \frac{\rho_m b}{\sqrt{\rho_f} \dot{\gamma}},$
athermal:	$\tau_a = \alpha \mu b \sqrt{C/C_0} (\sqrt{\rho_f} + \Lambda_b),$
lattice:	$\dot{\gamma} = \rho_m b \frac{\tau^* b}{B_d} \exp \left\{ -\frac{\Delta H_0}{2k_B T} \max \left[(1 - (\tau^*/\tau_P)^p)^q - \frac{T}{T_0}, 0 \right] \right\} \Rightarrow \tau^*,$
particles:	$\dot{\gamma} = \rho_m b \lambda_p [\nu_{\text{th}}(\tau_p) + \nu_{\text{cl}}(\tau_p)] \Rightarrow \tau_p,$
stress:	$\tau = \tau_a + \tau^* + \tau_p,$
at constant stress:	$\dot{\gamma}_{\text{tot}} = \dot{\gamma}_{\text{disl}}(\tau) + \dot{\gamma}_{\text{diff}}(\tau).$

7.1 Uniqueness

Every state variable, ρ , C/C_0 and ρ_m , is determined by $(\dot{\gamma}, T)$ and not by τ . For a given rate, therefore:

- the right-hand side of the lattice equation is strictly increasing in τ^* , so there is one τ^* ;
- the right-hand side of the particle equation is non-decreasing in τ_p , so there is one τ_p ;
- τ_a is a number.

The stress $\tau(\dot{\gamma})$ is therefore single valued. It need not be monotonic in $\dot{\gamma}$: where aging dominates it decreases with rate ($m < 0$), but it never has more than one value for a given rate.

7.2 Why the state is not written as a function of the stress

One could instead let the athermal stress depend on the stress, for example $\tau_a = \tau_{a0}(T)[1 - s_d \dot{\gamma}_{\text{diff}}(\tau)/\dot{\gamma} - \dots]$, which makes the equation for τ implicit and coupled. That closure is mathematically admissible, since $\tau - \tau_a(\tau)$ still increases with τ , but it has three drawbacks.

1. The softening coefficients are phenomenological; nothing links the share of strain carried by another mechanism to the strength of the forest.
2. A process that already carries strain in parallel would also soften the glide resistance, so its effect is counted twice.

3. The derivative $d\tau/d\ln\dot{\gamma}$ acquires a feedback factor $1/(1 - \partial\tau_a/\partial\tau)$ that mixes mobility and resistance, so the decomposition of Sec. 8 is no longer clean.

In the present rule the rate dependence of τ_a is still there, $\tau_a = \tau_a(\dot{\gamma}, T)$, but it comes from the kinetics of ρ and C/C_0 . A stress dependence could be added in a physically based way, for instance through the climb mobility $\propto \exp(\sigma b^3/k_B T)$ used by Sandström [20]. Uniqueness would then have to be checked explicitly.

8 Strain-rate sensitivity: exact decomposition

8.1 Derivation

Write $\tau = \tau^* + R$, where $R = \tau_a + \tau_p$ collects every resistance, and $\dot{\gamma} = G(\tau^*; \mathbf{s}) = \rho_m b v_g(\tau^*)$. Differentiating along the solution,

$$d\ln\dot{\gamma} = A d\tau^* + \frac{\partial \ln G}{\partial \mathbf{s}} \cdot d\mathbf{s}, \quad A = \frac{\partial \ln G}{\partial \tau^*} = \frac{\partial \ln v_g}{\partial \tau^*}. \quad (8.1)$$

Solving for $d\tau^*$ and adding dR :

$$\frac{d\tau}{d\ln\dot{\gamma}} = \frac{1}{A} + \underbrace{\frac{dR}{d\ln\dot{\gamma}} - \frac{1}{A} \frac{\partial \ln G}{\partial \mathbf{s}} \cdot \frac{d\mathbf{s}}{d\ln\dot{\gamma}}}_B. \quad (8.2)$$

Define $\Lambda \equiv AB$. Then

$$\boxed{m = \frac{1}{\tau} \frac{d\tau}{d\ln\dot{\gamma}} = \frac{1 + \Lambda}{\tau A}, \quad V^* = \frac{k_B T}{m\tau} = \frac{k_B T A}{1 + \Lambda}} \quad (8.3)$$

In practice A is evaluated from Eq. (2.3) at the solved τ^* , $d\tau/d\ln\dot{\gamma}$ is obtained by differentiating the full solution, and $B = d\tau/d\ln\dot{\gamma} - 1/A$.

8.2 Contributions to B

Source	Mechanism	Sign of B
Climb recovery	ρ falls as $\dot{\gamma}$ falls, Eq. (4.5)	+
Particle climb and detachment	$d\tau_p/d\ln\dot{\gamma} = 1/A_p$	+
Solute aging	C/C_0 falls as $\dot{\gamma}$ rises, Eq. (5.4)	−
Dynamic recovery, storage	rate independent	0

8.3 Limits and consequences

- *Mobility control* ($|\Lambda| \ll 1$): $m = 1/(\tau A)$ and $V^* = k_B T A$. Writing $m = (\tau^*/\tau) \cdot 1/(\tau^* A)$ shows that m can rise and then fall with temperature within a single mechanism: $1/(\tau^* A)$ grows as the barrier softens, while $\tau^*/\tau \rightarrow 0$ on the athermal plateau. V^* , in contrast, increases monotonically.
- *Resistance control* ($\Lambda \gg 1$): $m \simeq B/\tau$ and $V^* \simeq k_B T/B$. In the recovery limit Eq. (4.7), $B = \tau_a/3$ and $m \rightarrow 1/3$.

- *Negative rate sensitivity:*

$$m < 0 \iff 1 + \Lambda < 0 \iff \Lambda < -1, \quad (8.4)$$

which requires $B < 0$, i.e. aging, with $|B| > 1/A$. Under strain-rate control this branch is stable; under stress control it is not, and the rate jumps.

- A maximum in $V^*(T)$ marks the passage from A - to B -control; a maximum in $m(T)$ alone does not.

9 How the equations are solved

9.1 Imposed rate (tension)

The rate $\dot{\gamma}$ is known, and the steps are carried out in order:

1. compute $c_3 = KD/\dot{\gamma}$ and solve Eq. (4.5) for x by bisection, giving ρ , ρ_f and ρ_m ;
2. compute t_a from Eq. (5.1) and C/C_0 from Eq. (5.2);
3. evaluate τ_a from Eq. (4.1);
4. solve $\dot{\gamma} = \rho_m b v_g(\tau^*)$ for τ^* by bisection in $\ln \tau^*$;
5. solve Eq. (3.7) for τ_p by bisection;
6. add: $\tau = \tau_a + \tau^* + \tau_p$.

Each step uses only results of earlier steps. There are three scalar root finds, each on a monotonic function, and no iteration between them.

9.2 Imposed stress (creep, mechanism maps)

The rate is now the unknown, and since the state depends on it the problem is implicit in $\dot{\gamma}$:

$$F(\dot{\gamma}) \equiv \tau_a(\mathbf{s}(\dot{\gamma})) + \tau^*(\dot{\gamma}, \mathbf{s}(\dot{\gamma})) + \tau_p(\dot{\gamma}) - \tau_{\text{applied}} = 0. \quad (9.1)$$

For each temperature, $\tau(\dot{\gamma})$ is tabulated on $10^{-14} \leq \dot{\gamma} \leq 10^6 \text{ s}^{-1}$ using the rate-controlled solution and inverted at τ_{applied} . If $\tau(\dot{\gamma})$ is non-monotonic (DSA), the lowest-rate branch that reaches the stress is taken. The diffusional rate, Eq. (6.1), is then added.

9.3 Transients

The steady state is appropriate for flow stresses at saturation and for minimum creep rates. For transients (rate jumps, relaxation, primary creep), Eq. (4.4) and the aging time are integrated in time as internal variables. Each step is then explicit: from the current τ and state the flow rule gives $\dot{\gamma}$, and the state is updated. No $1/\dot{\gamma}$ appears in the time form.

10 Complete equations for each material

Conventions: $\dot{\gamma} = \dot{\epsilon}/M_s$ and $\tau = M_s\sigma$, with $M_s = 0.333$ for W and Eurofer-97 and 0.45 for Zr. Energies are in eV, $k_B = 8.617 \times 10^{-5}$ eV/K.

10.1 Level I: tungsten

W: kink-pair glide against a Kocks–Mecking forest

Here $\beta = f_m = 1$, so $\rho_f = \rho_m = \rho$, and $\Lambda_b = 0$. There is no aging and there are no particles. Given $(\dot{\gamma}, T)$:

$$k_1\sqrt{\rho} - k_{20}e^{q_2/k_BT}\rho - \frac{K D_0 e^{-Q_D/k_BT}}{\dot{\gamma}} \rho^2 = 0, \quad (10.1)$$

$$\dot{\gamma} = \rho b \frac{\tau^* b}{B_d} \exp \left\{ -\frac{\Delta H_0}{2k_BT} \max \left[\left(1 - \left(\frac{\tau^*}{\tau_P} \right)^p \right)^q - \frac{T}{T_0}, 0 \right] \right\}, \quad (10.2)$$

$$\boxed{\tau = \alpha \mu b \sqrt{\rho} + \tau^*} \quad (10.3)$$

With the self-diffusivity of W ($Q_D = 6.0$ eV) the climb term is negligible below ~ 1000 K, so $\sqrt{\rho} = k_1/k_2(T)$ and

$$\tau \simeq \alpha \mu b \frac{k_1}{k_{20}} e^{-q_2/k_BT} + \tau^*(\dot{\gamma}, T). \quad (10.4)$$

Check: at 300 K, $k_2 = 500 e^{-0.011/0.02585} = 327$, $k_1/k_2 = 1.68 \times 10^7 \text{ m}^{-1}$, and $\tau_a = 0.28 \times 161 \text{ GPa} \times 0.2722 \text{ nm} \times 1.68 \times 10^7 \text{ m}^{-1} = 207 \text{ MPa}$.

b, μ, T_m, Ω	0.2722 nm, 161 GPa (constant), 3695 K, $1.585 \times 10^{-29} \text{ m}^3$
$\Delta H_0, \tau_P, p, q$	2.4 eV, 0.8 GPa, 0.83, 1.69
T_0, B_d	$0.9T_m = 3326$ K, 1.10×10^{-3} Pa s
α, k_1, k_{20}, q_2	0.28, $5.5 \times 10^9 \text{ m}^{-1}$, 500, -0.011 eV
K, D_0, Q_D	10^9 , $4 \times 10^{-5} \text{ m}^2/\text{s}$, 6.0 eV (not calibrated)
creep: d, D_{b0}, Q_b	100 μm , $4 \times 10^{-5} \text{ m}^2/\text{s}$, 3.9 eV

10.2 Level II: zirconium (prismatic slip)

Zr: level I + oxygen aging + climb recovery

Here $\beta = f_m = 1$ and $\Lambda_b = 0$. With $\rho_m = \rho_f = \rho$ the waiting time simplifies to $t_a = b\sqrt{\rho}/\dot{\gamma}$. Given $(\dot{\gamma}, T)$:

$$k_1\sqrt{\rho} - k_{20}e^{q_2/k_B T}\rho - \frac{K D_0 e^{-Q_D/k_B T}}{\dot{\gamma}} \rho^2 = 0, \quad (10.5)$$

$$\frac{C}{C_0} = 1 + C_O \left\{ 1 - \exp \left[- \left(\frac{b\sqrt{\rho}}{\dot{\gamma}} \nu_O e^{-Q_O/k_B T} \right)^{\alpha_O} \right] \right\}, \quad (10.6)$$

$$\dot{\gamma} = \rho b \frac{\tau^* b}{B_d(\Delta g)} \exp \left\{ - \frac{\Delta H_0}{2k_B T} \Delta g \right\}, \quad \Delta g = \max \left[\left(1 - \left(\frac{\tau^*}{\tau_P} \right)^p \right)^q - \frac{T}{T_0}, 0 \right], \quad (10.7)$$

$$\tau = \alpha \mu b \sqrt{\frac{C}{C_0}} \sqrt{\rho} + \tau^* \quad (10.8)$$

with $B_d(\Delta g)$ from Eq. (2.2). For creep, $\dot{\gamma}_{\text{tot}} = \dot{\gamma}_{\text{disl}} + 42 \Omega \tau (D_v + \pi \delta D_b/d)/(k_B T d^2)$.

What each addition does. Aging raises the plateau above ~ 600 K and makes it decrease with rate ($\Lambda < -1$ near 650 K). Climb recovery makes ρ , and hence τ_a , fall steeply above ~ 750 K at low rates, approaching $\dot{\gamma} \propto D\tau^3$.

b, μ, T_m, Ω	0.3233 nm, 33 GPa (constant), 2128 K, $2.33 \times 10^{-29} \text{ m}^3$
$\Delta H_0, \tau_P, p, q$	3.75 eV, 300 MPa, 0.86, 1.69
T_0, B_k, B_0	$0.7T_m = 1490$ K, 7.65×10^{-4} Pa s, 5.0×10^{-5} Pa s
α, k_1, k_{20}, q_2	0.36, $5.5 \times 10^9 \text{ m}^{-1}$, 500, -0.005 eV
O aging: $C_O, Q_O, \nu_O, \alpha_O$	0.30, 1.80 eV, 10^{13} s^{-1} , 2/3
recovery: K, D_0, Q_D	$10^9, 10^{-4} \text{ m}^2/\text{s}$, 3.2 eV
creep: d, D_v, D_{b0}, Q_b	$20 \text{ }\mu\text{m}, = D, 10^{-4} \text{ m}^2/\text{s}$, 2.1 eV

10.3 Level III: Eurofer-97

Eurofer-97: level II + boundaries + particles + diffusional flow

Here $\beta = 0.1$ and $f_m = 1$, so $\rho_f = \rho_m = 0.1\rho$, and $t_a = b\sqrt{\rho_f}/\dot{\gamma}$. The shear modulus is temperature dependent, $\mu(T) = \mu_0 \max\{1 - 0.35(T - 300)/(T_m - 300), 0.45\}$ with $\mu_0 = 80$ GPa. Given $(\dot{\gamma}, T)$:

$$k_1(\sqrt{\rho} + \Lambda_b) - k_{20}e^{q_2/k_B T} \rho - \frac{K D_0 e^{-Q_D/k_B T}}{\dot{\gamma}} \rho^2 = 0, \quad (10.9)$$

$$\frac{C}{C_0} = 1 + C_{\text{CN}} \left\{ 1 - \exp \left[- \left(\frac{b\sqrt{\rho_f}}{\dot{\gamma}} \nu_{\text{CN}} e^{-Q_{\text{CN}}/k_B T} \right)^{\alpha_{\text{CN}}} \right] \right\}, \quad (10.10)$$

$$\tau_a = \alpha\mu(T)b\sqrt{\frac{C}{C_0}}(\sqrt{\rho_f} + \Lambda_b), \quad (10.11)$$

$$\dot{\gamma} = \rho_f b \frac{\tau^* b}{B_d} \exp \left\{ - \frac{\Delta H_0}{2k_B T} \max \left[\left(1 - \left(\frac{\tau^*}{\tau_P} \right)^p \right)^q - \frac{T}{T_0}, 0 \right] \right\}, \quad (10.12)$$

$$\dot{\gamma} = \rho_f b \lambda_p \left[\nu_0 \exp \left\{ - \frac{F(T)}{k_B T} \left(1 - \frac{\tau_p}{\hat{\tau}(T)} \right) \right\} + \frac{2\pi D_c}{b h \ln(R/r_c)} \frac{e^{\tau_p \Omega/k_B T} - 1}{\max(1 - \tau_p/\hat{\tau}(T), 10^{-3})} \right], \quad (10.13)$$

$$\tau = \alpha\mu(T)b\sqrt{\frac{C}{C_0}}(\sqrt{\rho_f} + \Lambda_b) + \tau^* + \tau_p \quad (10.14)$$

$$\text{creep: } \dot{\epsilon} = M_s \left[\dot{\gamma}_{\text{disl}}(M_s \sigma) + 42 \frac{\Omega M_s \sigma}{k_B T d^2} \left(D + \frac{2\pi b}{d} D_b \right) \right], \quad (10.15)$$

with $\hat{\tau}(T) = \hat{\tau}_0 \mu(T)/\mu_0$, $F(T) = F_0 \mu(T)/\mu_0$ (detachment with $p = q = 1$), $D_c = D_{c0} e^{-Q_c/k_B T}$, and $Q_c = Q_D$.

The boundary term collects laths, blocks, prior-austenite grains and particles:

$$\Lambda_b = \frac{1}{0.5 \mu\text{m}} + \frac{0.3}{3.1 \mu\text{m}} + \frac{0.1}{21 \mu\text{m}} + \frac{0.3}{1.5 \mu\text{m}} + \frac{0.2}{0.456 \mu\text{m}} = 2.74 \times 10^6 \text{ m}^{-1}. \quad (10.16)$$

What each addition does. Λ_b raises storage and the plateau. C/N aging gives the flattened m near 400–700 K. The particles carry more than half of the flow stress; their detachment makes τ_p fall with temperature, and their local climb produces threshold creep with $n \approx 16$ –18 at 450–500 °C and 5–8 at 650 °C. Diffusional flow sets the floor of the creep rate at low stress.

b, μ_0, T_m, Ω	0.2737 nm, 80 GPa, 1811 K, $1.18 \times 10^{-29} \text{ m}^3$
$\Delta H_0, \tau_P, p, q$	1.493 eV, 680.6 MPa, 0.65, 1.7
T_0, B_d	$0.9T_m = 1630 \text{ K}$, $8.75 \times 10^{-4} \text{ Pa s}$
$\alpha, k_1, k_{20}, q_2, \beta$	0.2959, $7.8 \times 10^{10} \text{ m}^{-1}$, 6750, -0.007 eV , 0.1
Λ_b	$2.74 \times 10^6 \text{ m}^{-1}$
C/N aging: C, Q, ν, α	1.206, 0.965 eV, 10^{13} s^{-1} , 1/3
recovery: K, D_0, Q_D	6.59×10^6 , $2 \times 10^{-4} \text{ m}^2/\text{s}$, 2.917 eV
particles: $\hat{\tau}_0, F_0, \nu_0, \lambda_p$	188.4 MPa, 3.658 eV, 10^{11} s^{-1} , 200 nm
particle climb: $D_{c0}, Q_c, h, \ln(R/r_c)$	$5.365 \times 10^{-3} \text{ m}^2/\text{s}^\dagger$, 2.917 eV, 1 nm, 5
diffusional: d, D_{b0}, Q_b, δ	21 μm , $2 \times 10^{-4} \text{ m}^2/\text{s}$, 1.8 eV, 2b

[†]Lumped with h and $\ln(R/r_c)$; not a bare diffusivity.

11 Summary of the hierarchy

Ingredient	W (I)	Zr (II)	Eurofer-97 (III)
Orowan + kink-pair travel, Eq. (2.1)	✓	✓	✓
Taylor stress, forest storage, dynamic recovery	✓	✓	✓
Climb (static) recovery, $KD\rho^2/\dot{\gamma}$	(inactive)	✓	✓
Solute aging, $\sqrt{C/C_0}$	–	O	C, N
Boundary storage, Λ_b	–	–	✓
Particle group: detachment climb	–	–	✓
Diffusional flow (creep)	prediction	prediction	✓
Sign of B	0	– then +	– then +

Each level keeps the equations of the level below unchanged and adds terms at fixed places: new resistances enter τ , new rate-dependent state enters $\mathbf{s}(\dot{\gamma}, T)$, and new independent carriers enter $\dot{\gamma}_{\text{tot}}$.

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